MASTER OPERATIONAL BIBLE • VERSION 5.0 PRODUCTION EDITION

Complete Factory Operations, User Interface Navigation & Loss Control Manual

The Exhaustive Guide for Factory Owners, Supervisors, Cashiers & Drivers

Executive Operational Mandate: This document serves as the complete, permanent training and operating bible for the Obi Table Water & Bottling Plant. It covers every screen, every button, every financial calculation, and every anti-theft protocol enforced by the Obi Table Water Operating System.

📖 TABLE OF CONTENTS

Section 1: Executive Overview & The Dual-Line Manufacturing Model — Page 2
Section 2: Complete User Interface & Screen-by-Screen Navigation Guide — Page 3
• Screen 1: Staff Visual Login & PIN Authentication (/factory/login) — Page 3
• Screen 2: Supervisor Console — Dual Production Lines (Kenneth) — Page 4
• Screen 3: Gate Cashier POS & Moniepoint RRN Lock (Blessing) — Page 5
• Screen 4: Route Delivery Driver Manifest & Jar Quality (Musa) — Page 6
• Screen 5: Executive Owner Portal & Financial P&L (Mr. Obi) — Page 7
• Screen 6: Interactive Video Training Center (/factory/walkthrough) — Page 8
Section 3: General Factory Operations Manual & Water Treatment Protocol — Page 9
Section 4: The 5 Anti-Fraud Standard Operating Procedures (SOPs) — Page 10
Section 5: Daily Shift Timetable & Machine Maintenance Protocols — Page 11
Section 6: Emergency Troubleshooting, Offline Recovery & Factory Contact — Page 12

Plant Identification & Technical Support Registry:

- **Factory Name:** Obi Table & Bottled Water Plant
- **Operating System:** Obi Table Water OS (v5.0 Hardened)
- **Production Lines:** Sachet Water + Bottled (50cl/75cl) + 19L Jars
- **Corporate Desk:** Bethelmind Digital Solutions (CAC: 9912717)
- **Owner Desk:** wa.me/2348022791227 (Hotline: 0802 279 1227)
- **Corporate Bank:** Moniepoint Microfinance Bank
- **Settlement Account:** 6805375225 (Bethelmind Digital Solutions)
- **Cloud Hosting:** Vercel Production Edge • Offline-Ready

1. Executive Overview & The Dual-Line Manufacturing Model

In the Nigerian commercial water industry, factories that rely exclusively on sachet water ("pure water") operate on razor-thin margins of ₦25 to ₦40 per bag. A single generator breakdown, an unrecorded driver shortage, or a sudden spike in diesel prices immediately wipes out weekly profits.

Mr. Obi's plant operates on a **Dual-Line Hybrid Model** that pairs the high turnover of sachet water with the high contribution margins of PET bottled water and 19-Liter dispenser jar refills:

Line & Product	Packaging Spec	Wholesale Price	Unit Direct Cost	Net Unit Margin	Turnover Speed	Primary Vulnerability
Line 1: Sachet Pure Water	Bag of 20 sachets (500ml)	₦350 / bag	₦310 / bag	**₦40 / bag (11.4%)**	Extremely Fast (4,200 bags/day)	Driver road burst claims; unrecorded ghost shifts.
Line 2: 75cl Bottled Water	Shrink pack of 12 bottles	₦1,200 / pack	₦920 / pack	**₦280 / pack (23.3%)**	Fast (250 packs/day)	Preform theft masked as "blow-mold rejects".
Line 2: 50cl Bottled Water	Shrink pack of 20 bottles	₦1,100 / pack	₦850 / pack	**₦250 / pack (22.7%)**	Fast (150 packs/day)	Cashier unrecorded sales; label shrinkage.
Line 2: 19L Dispenser Jars	19-Liter Refill (Polycarbonate)	₦900 / refill	₦320 / refill	**₦580 / jar (64.4%)**	Moderate (40–100 jars/day)	**Unreturned or battered empty jars (₦4,500 ea).**

1.2 Daily Revenue & Contribution Split (Standard 1-Shift Day)

💧 Sachet Line Contribution
- Volume: 4,200 Bags (14 Nylon Rolls)
- Gross Sales: **₦1,470,000** (74.6% of Plant Revenue)
- Consumables: 280kg LDPE film (₦280,000)
- Net Cash Contribution: **₦168,000**

🍾 Bottling & Dispenser Line Contribution
- Volume: 400 packs + 40 dispenser jars
- Gross Sales: **₦501,000** (25.4% of Plant Revenue)
- Consumables: 3,000 preforms, caps, shrink film (₦123,600)
- Net Cash Contribution: **₦130,700**

1.3 The Core Operational Formula

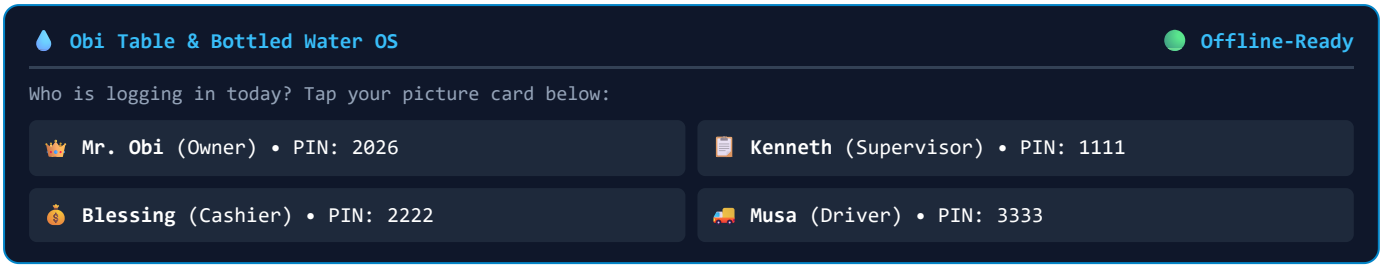
Daily Plant Health Equation: $$ ext{Today's Net Cash Profit} = ext{Total Gross Gate Sales} - (ext{Diesel Cost} + ext{Packaging Consumables} + ext{Labour Wages} + ext{Driver Shortage Debits})$$ On this system, Mr. Obi never waits for monthly account reconciliation. At 8:00 PM every evening, his mobile screen reflects the confirmed Net Cash in hand: **₦524,200**.

2. Screen-by-Screen UI Navigation Guide

Screen 1: Zero-Password Staff Visual Login

URL: `https://www.bethelmindanalytics.com/factory/login` • **Access:** All Factory Personnel & Plant Owner

Traditional email and password systems fail catastrophically in Nigerian manufacturing environments. Machine operators and drivers forget passwords, share credentials, or lock accounts. Obi Table Water OS replaces passwords with a high-contrast, visual role selector and 4-digit numeric PINs.



Navigation & Interaction Checklist:

1. **Selecting Your Role:** Tap on the large rectangular card bearing your name and icon. The role selector smoothly slides into the 4-digit PIN pad.
2. **Entering Your PIN:** A 3×4 touch keypad is displayed with large numbered buttons (1 to 9 , 0 , and Clear). Tap your 4-digit PIN. As you tap each digit, the 4 PIN dots fill with blue illumination.
3. **Server-Side Constant-Time Validation:** Upon tapping the 4th digit, the system dispatches a background cryptographic request to `/api/factory/auth` . A blue status indicator states: "*Verifying PIN with Server Security...*".
4. **Incorrect PIN Handling:** If the wrong PIN is entered, the 4 dots pulse red, the terminal vibrates, and a clear red error message appears: "*Incorrect PIN entered. 4 attempts remaining.*" The keypad resets automatically after 1.2 seconds.
5. **Brute-Force Rate Limiter:** If an unauthorized person enters 5 incorrect PINs consecutively, the security engine blocks the IP for 15 minutes, preventing dictionary attacks.
6. **Offline Fallback:** If factory Wi-Fi or cellular broadband drops completely, the system seamlessly verifies the PIN against encrypted local storage so factory operations never freeze.
7. **Switching Staff:** To log out or hand over to another worker, tap the red Switch Staff button in the top right corner.

> **Training Tip for New Staff:** During onboarding, workers can tap the green text link at the bottom: **"Auto-Enter Demo PIN →"** to immediately understand how the console opens without typing.

Screen 2: Shift Supervisor Dual-Line Console (Kenneth)

Role: Kenneth (Plant Supervisor) • **PIN:** `1111` • **Shift Time:** 7:00 AM – 6:30 PM

Kenneth is responsible for machine outputs, nylon consumption, preform blow-molding quality, generator fuel dips, and dispatching the official evening shift summary to Mr. Obi.

```
Kenneth • Dual-Line Production Supervisor                    Batch A Live

[ 1. Sachet Water Line ]  [ 2. Bottled Water & Jars ]

Sachet Nylon Rolls Used: 14 rolls (= 4,200 bags packed • Value: ₦1,470,000)
[ + Finished Another Roll (+1 Roll) ]  [ -1 Roll ]

⚖️ Preform Rejects & Scrap Scale Audit: 24 pcs (= 432g target)
Digital Scale Weight: 432 grams ✓ Weight Verified

Generator Fuel Poured: [20L] [25L] [30L] [40L]
```

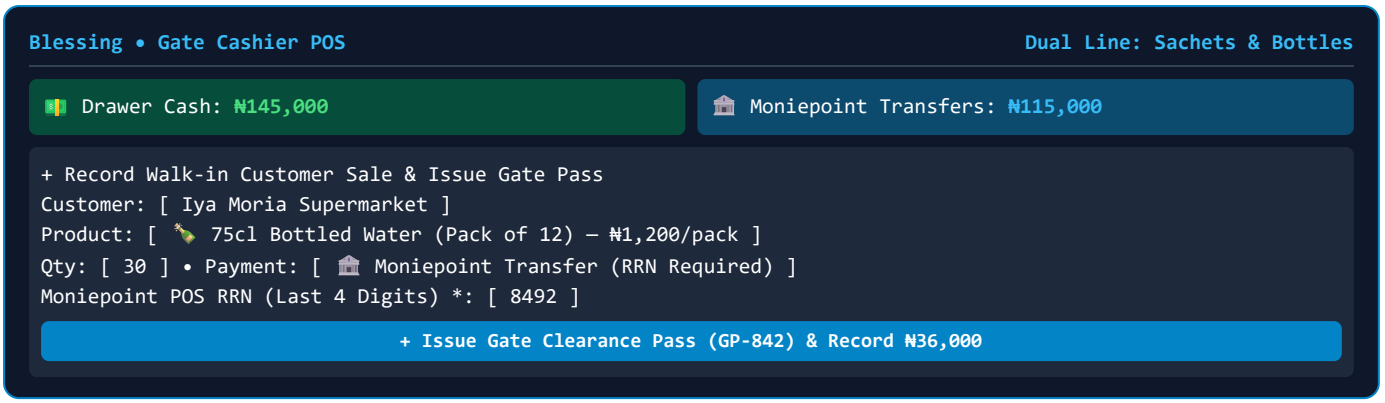
Navigation & Operational Controls:

1. **Production Line Tabs:** Tap **1. Sachet Water Line** to view and log pure water bags. Tap **2. Bottled Water & Jars** to view 75cl, 50cl packs, 19L dispenser jars, and preform audits.
2. **Logging Sachet Production (+1 Roll Used):** Whenever a 20kg printed LDPE nylon roll finishes on the sealing machine, Kenneth taps the large green **+ Finished Another Roll (+1 Roll)** button. The system automatically increments the roll count, adds 300 bags to output, updates total value to ₦1,470,000, and auto-saves to local storage.
3. **Correcting Sachet Errors (-1 Roll):** If tapped by mistake, Kenneth taps the grey **-** button to decrement.
4. **Logging Bottled Water Production:** In Tab 2, Kenneth taps **+10 Packs** for 75cl bottles (₦1,200/pack), **+10 Packs** for 50cl bottles (₦1,100/pack), and **+5 Jars** for 19L dispenser refills (₦900/jar).
5. **PET Preform Blow Rejects & Digital Scrap Scale Audit:**
   - Kenneth enters the number of claimed defective bottles in the "*Claimed Blow Rejects*" field (e.g. 24 pcs).
   - The system calculates the theoretical weight: $24 imes 18.0 ext{g} = 432 ext{ grams}$.
   - Kenneth crushes the plastic rejects, places them on the physical digital scale, and types the scale reading into "*Digital Scale Weight*".
   - If scale matches within ±15g: Green pill displays **✓ WEIGHT VERIFIED**.
   - If scale is deficient (e.g. 200g): Red alert triggers: "⚠️ *Preform Smuggling Risk: Missing 232g of PET plastic!*"
6. **Logging Generator Fuel Poured:** Tap one of the 4 fuel buttons: **20L**, **25L**, **30L**, or **40L** based on liters poured from the drum.
7. **Sending Evening Shift Report to Mr. Obi:** At 6:00 PM, Kenneth taps the green WhatsApp bar: **📲 Send Dual-Line Shift Close to Mr. Obi on WhatsApp**. This compiles all sachet rolls, bottle packs, dispenser jars, preform scale balance, and diesel dips into a clean formatted text report and opens WhatsApp with Mr. Obi's direct line (`0802 279 1227`) pre-filled.

Screen 3: Gate Cashier POS & Moniepoint RRN Lock (Blessing)

Role: Blessing (Gate Cashier) • **PIN:** `2222` • **Location:** Factory Gate Sales Kiosk

Blessing is responsible for issuing receipts to walk-in distributors, verifying bank transfers, preventing fake alerts, and generating Gate Clearance Passes.



Navigation & POS Transaction SOP:

1. **Drawer Cash vs. POS Transfer Reconciliation:** Top cards display live totals for physical paper notes collected vs. confirmed electronic bank transfers. Physical drawer cash must balance to the exact Naira at shift closing.
2. **Recording a Customer Sale:**
   - Type customer/depot name in "*Customer / Depot Name*" (e.g., Madam Chichi Stores).
   - Select product from the dropdown:
     - 💧 Sachet Water (Bag of 20) — ₦350/bag
     - 🍾 75cl Bottled Water (Pack of 12) — ₦1,200/pack
     - 🍡 50cl Bottled Water (Pack of 20) — ₦1,100/pack
     - 🛢️ 19L Dispenser Jar Refill — ₦900/jar
   - Enter Quantity (e.g., 30 packs).
   - Select Payment Method: `💵 Cash in Hand` or `🏦 Moniepoint Transfer (RRN Required)`.
3. **Moniepoint POS Transfer Reference (RRN) Lock:**
   - If *Cash* is selected: Blessing collects paper notes and clicks *Record Sale* immediately.
   - If *Moniepoint Transfer* is selected: The system reveals a mandatory security field: "*Moniepoint POS Terminal RRN (Last 4 Digits)* *".
   - Blessing MUST inspect the printed Moniepoint POS transaction slip and type the last 4 digits of the Retrieval Reference Number (e.g. **8492**).
   - If left blank, the system displays an alert and blocks receipt generation.
4. **Gate Clearance Pass Code (GP-xxx):** Upon successful recording, the software generates a 3-digit clearance code (e.g. `GP-842`). This code is logged in the register and quoted to gate security.
5. **Gate Security Inspection:** Factory security guards will **NOT** open the gate boom barrier for the distributor vehicle until the driver recites the matching Gate Pass code verified against Blessing's console.

Screen 4: Route Delivery Driver Manifest & Jar Quality (Musa)

Role: Musa (Truck 2 Driver) • **PIN:** `3333` • **Route:** Badore / Ajah Corridor

Musa is responsible for delivering mixed loads, accounting for vehicle cash collections, returning empty plastic pouches for burst claims, and inspecting empty 19L jars.



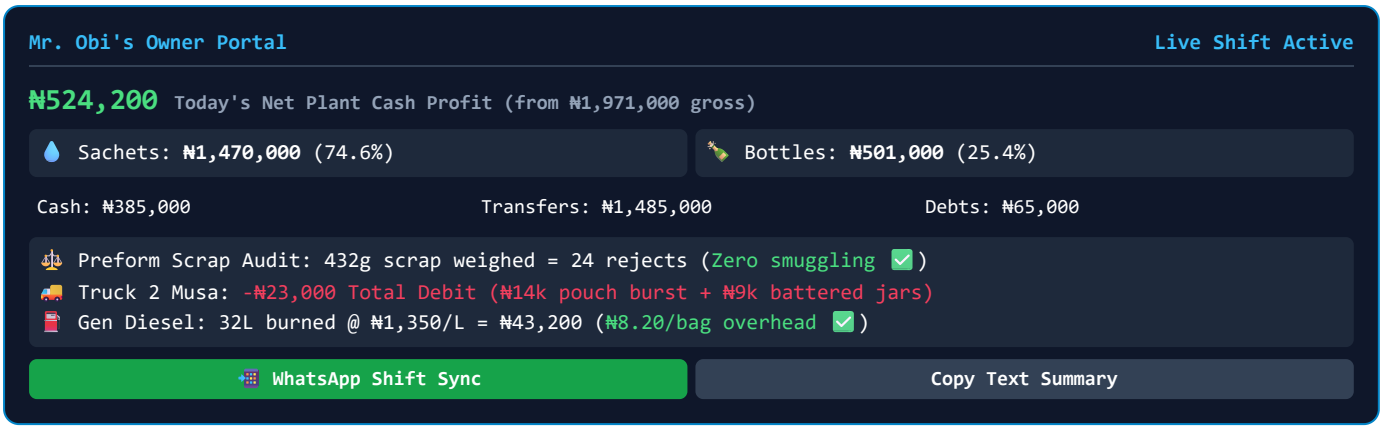
Navigation & Driver Reconciliation Rules:

1. **Morning Dispatch Verification:** Before leaving the loading bay, Musa opens his console and confirms the physical inventory loaded on Truck 2 matches the screen: 700 sachet bags, 60 bottle packs, and 15 dispenser jars. Total remittance expected: ₦330,500.
2. **Evening Handover & Settlement:** Musa submits collected physical cash and Moniepoint POS customer receipts to Blessing at the cashier window.
3. **The Burst Pouch Rule (Sachet Water):**
   - If Musa claims 40 bags burst due to bad potholes along Badore road, he must deliver **40 physical wet burst plastic pouches** to Kenneth at the gate.
   - Kenneth counts the empty pouches. If Musa delivers **0 pouches**, the claim is rejected as driver pocketing.
   - The system automatically logs an auto-debit of $40 imes ext{₦350} = mathbf{₦14,000}$ against Musa's account.
4. **19L Dispenser Jar Quality Inspection:**
   - For the 15 refilled jars dispatched, Musa must return 15 usable empty jars.
   - Gate loaders inspect the returned containers: clean Obi-branded jars are accepted.
   - If Musa brings back cracked, battered, or generic third-party competitor jars (e.g. 2 battered jars), they are marked in the "*Battered / Off-Brand Jars*" field.
   - The system immediately calculates a replacement debit: $2 imes ext{₦4,500} = mathbf{₦9,000}$, debited from Musa's weekly commission.

Screen 5: Executive Owner Portal & Financial P&L (Mr. Obi)

Role: Mr. Obi (Managing Director & Plant Owner) • **PIN:** `2026` • **URL:** `/factory/owner`

Mr. Obi's portal is the master executive control tower. Accessible directly from his smartphone home screen, it delivers real-time visibility into net cash profits, inventory security, fuel overhead, and fleet shortages.



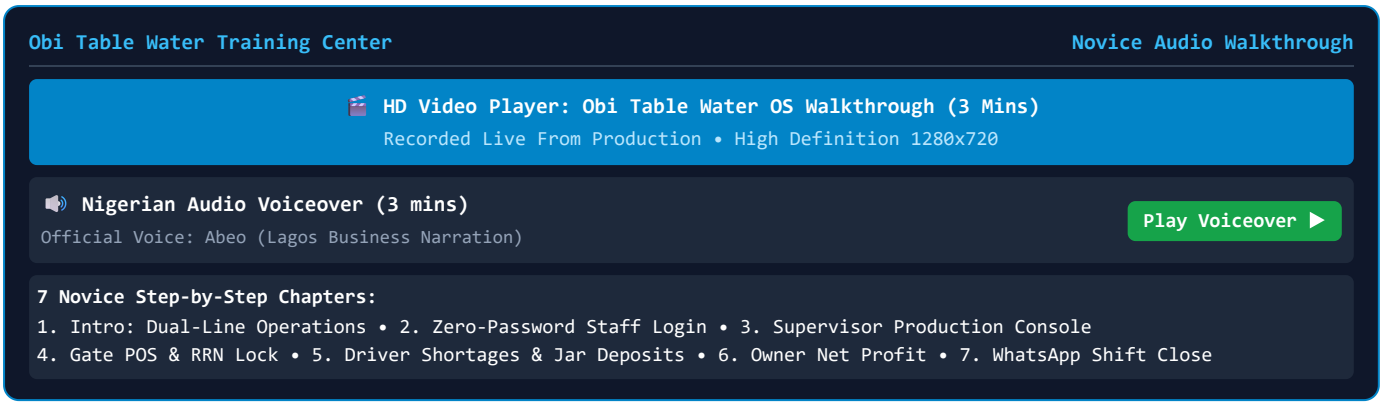
Navigation & Executive Oversight Checklist:

1. **4-Digit Owner Authentication:** Visit `/factory/owner`. Enter PIN `2026`. The executive dashboard paints instantaneously.
2. **Net Cash Profit Hero Card:** The top banner displays the real-time net cash contribution: **₦524,200**. It details the revenue split: 74.6% Sachet Line (₦1,470,000) vs 25.4% Bottled Line (₦501,000).
3. **Vault Breakdown:** Displays physical paper cash in Blessing's drawer (₦385,000), confirmed Moniepoint POS transfers (₦1,485,000), and outstanding distributor debts (₦65,000).
4. **Preform Grammage Audit Verification:** Confirms that the physical scale weight of crushed reject plastic (432g) matches the theoretical weight of 24 reported blow rejects, ensuring zero preform theft.
5. **Fleet Shortage & Jar Liability Card:**
   - Truck 1 (Emeka): Balanced (₦516,000 remitted ✅).
   - Truck 2 (Musa): Flagged for **-₦23,000 debit** (₦14,000 for 40 missing sachets with 0 pouches returned + ₦9,000 for 2 battered dispenser jars).
6. **Generator Diesel Overhead Slicer:** Monitors diesel consumption: 32 Liters burned over 4.0 hours @ ₦1,350/L = ₦43,200. Energy cost per sachet bag = **₦8.20/unit** (well below the ₦12.00/unit profitability threshold).
7. **NAFDAC Quality Log:** Permeate TDS: 24 ppm (< 100 ppm PASS ✅), pH: 7.2 (Neutral PASS ✅).
8. **1-Tap Phone Home Screen Access:** Tap the browser menu and select "*Add to Home Screen*" to turn the portal into a standalone native mobile app with an Obi Table Water icon.

Screen 6: Interactive Video Training Center & Voiceover Player

URL: `https://www.bethelmindanalytics.com/factory/walkthrough` • **Audience:** Factory Owner, New Staff, Onboarding Trainees

To eliminate staff confusion and training overhead, the system includes a complete multimedia training academy embedded directly in the web application.



Navigation & Features Guide:

1. **Embedded HD Video Player:** Tap the video play button to watch a 3-minute, high-definition walkthrough captured directly from the live software. Demonstrates all role logins, button clicks, and audit reconciliations.
2. **Synchronized Audio Voiceover:** Tap the **Play Voiceover ▶** button on the top controller bar to listen to the crystal-clear Nigerian English business narration (recorded by Abeo in Lagos).
3. **The 7 Novice Chapters:** Beneath the video player, seven visual lesson cards break down each screen step-by-step. Each card includes:
   - Exact video timestamp range (e.g. 00:55 - 01:30).
   - Staff role badge (Owner, Supervisor, Cashier, Driver).
   - Clear, non-technical beginner explanation.
   - Core Factory Rule (e.g., "*1kg of sachet nylon must yield 16 bags*").
   - **"Try Live" Action Button:** Tapping this button jumps directly into that live screen in the factory OS so staff can practice immediately.
4. **Direct Asset Downloads:**
   - Tap **Download Video (MP4 - 6.6 MB)** to save the full HD MP4 file directly to your smartphone for offline WhatsApp sharing or staff orientation.
   - Tap **Download Complete PDF Manual (431 KB)** to download this official printable manual.

3. General Factory Operations & Water Treatment Protocol

The foundation of a reputable, high-margin water manufacturing business in Nigeria is consistent water purity compliant with **NAFDAC Regulations & NIS 306:2008 Standards**.

3.1 The 6-Stage Industrial Water Treatment Train

Stage 1: Aeration & Sediment Settling
Raw borehole water is aerated into raw water tanks to precipitate dissolved ferrous iron (Fe^{2+}) into ferric hydroxide ($Fe(OH)_3$). Minimum settling time: 4 hours.

Stage 2: Manganese Greensand & Media
Multi-media sand filter removes turbidity and suspended particulate matter (> 20 microns). Backwashed every 48 operating hours.

Stage 3: Activated Granular Carbon (GAC)
High-iodine activated carbon absorbs dissolved organic matter, eliminates swampy borehole odors, and ensures crystal clarity and sweet taste.

Stage 4: Micro-Filtration Polishing (5µm / 1µm)
Dual polypropylene spun sediment cartridges trap micro-fines before water reaches high-pressure RO booster pumps. Cartridges replaced monthly.

Stage 5: Industrial Reverse Osmosis (RO)
High-pressure polyamide membrane rejects 98.5% of dissolved minerals, heavy metals, nitrates, and salts. Permeate TDS maintained at **15–45 ppm**.

Stage 6: Ozone Dosing & UV Sterilization
Water passes through an ultraviolet germicidal chamber (254nm wavelength) and mild ozone injection to provide residual shelf-life disinfection in sealed bottles.

3.2 Daily Laboratory Purity Log Standards

Laboratory Parameter	NAFDAC Standard	Obi Factory Safe Range	Testing Frequency	Corrective Action if Outside Range
Total Dissolved Solids (TDS)	< 100 ppm	**15 – 45 ppm**	Twice Daily (Morning & Noon)	Flush RO membrane with citric acid; check booster pump pressure.
Water pH (Acidity / Alkalinity)	6.5 – 8.5	**6.8 – 7.4 (Neutral)**	Twice Daily	Calibrate calcite re-mineralizer cartridge or adjust aeration dosing.
Electrical Conductivity (EC)	< 200 µS/cm	**30 – 90 µS/cm**	Daily	Inspect RO permeate seal rings for salt slippage.
UV Germicidal Lamp Hours	< 8,000 hrs	**< 7,500 hrs**	Logged Weekly	Replace UV quartz sleeve and lamp bulb upon reaching 7,500 hours.
Physical Taste & Odor	Odorless, Clean	**Sweet, 100% Pure**	Every Batch Start	Backwash activated carbon filter; replace carbon media if exhausted.

4. The 5 Anti-Fraud Standard Operating Procedures (SOPs)

These 5 mandatory procedures protect Mr. Obi's plant from the most common money leaks in Nigerian FMCG water operations:

SOP-01: Ghost Shift Prevention via Nylon Film Yield Audit

> **The Ghost Shift Money Leak:** Night-shift operators run machines after hours, seal 1,000 bags of water using factory power and water, sell them directly to local kiosks at ₦250/bag, and pocket ₦250,000 cash while claiming the machines were idle.

- **The Enforcement:** Every 20kg roll of printed LDPE film is serialized and weighed before issue. Standard roll yield is exactly **300 to 320 bags**.
- At the end of each shift, Kenneth must reconcile the total bags packed against the rolls consumed ($14 ext{ rolls} imes 300 = mathbf{4,200 ext{ bags}}$). Any discrepancy of more than 50 bags triggers an immediate investigation before staff depart.

SOP-02: Driver Shortage & Physical Pouch Return Enforcement

> **The Fake Pothole Burst Scam:** Route drivers sell 40 bags of sachet water into neighborhood kiosks, pocket ₦14,000 cash, and tell the cashier at evening: "*Bags burst on road because Badore road has terrible potholes.*"

- **The Enforcement:** Zero empty pouches = Zero burst credit.
- If Musa claims 40 bags burst, he must hand over **40 physical wet empty plastic pouches**.
- If he returns 0 pouches, the software auto-debits the full wholesale value ($40 imes ext{₦350} = mathbf{₦14,000}$) from his payout ledger.

SOP-03: PET Preform Blow Reject Scrap Scale Grammage Audit

> **Preform Siphoning Fraud:** Machine operators claim 50 bottles were ruined during blow-molding and smuggle 50 pristine, unblown PET preforms out of the factory to sell to cosmetic and chemical packagers for ₦40 each.

- **The Enforcement:** 1 standard 75cl PET preform weighs exactly **18.0 grams**.
- Kenneth must collect all deformed, rejected blown bottles, crush them, and place them on a digital gram scale.
- $ ext{Expected Scrap Weight} = ext{Claimed Rejects} imes 18.0 ext{g}$. For 24 rejects, the scale MUST read **432 grams** (±15g).
- If the scale reads only 200g, 232g of preforms (13 preforms) are missing. The system flags an alert and supervisor must account for the deficit.

SOP-04: Moniepoint POS Transfer Reference (RRN) Lock & Gate Pass

> **Cash Pocketing Collusion:** A cashier records a ₦36,000 cash sale as "Bank Transfer", pockets the ₦36,000 cash, and tells the owner at evening: "*The customer transferred money, but bank network delayed the SMS alert.*"

- **The Enforcement:** The software locks the transfer button. Blessing CANNOT record a transfer sale without entering the **last 4 digits of the POS slip RRN**.
- Upon entry, the software prints a 3-digit **Gate Clearance Pass (GP-xxx)**. Gate security will not open the boom barrier without this code.

SOP-05: 19L Dispenser Jar Quality Inspection & Battered Jar Liability

- An empty 19L polycarbonate dispenser jar costs **₦4,500 – ₦6,000**. When drivers accept battered, cracked, or third-party generic jars from customers, the factory absorbs a massive capital loss.

- Gate loaders inspect all returned empty jars. Battered or off-brand competitor containers are rejected at the gate and an automatic **₦4,500 replacement penalty** per damaged jar is debited to the driver's payout.

5. Daily Shift Timetable & Machine Maintenance Protocols

5.1 The Standard Factory Daily Routine (Morning to Night)

Time	Department / Role	Operational Activity	System Action on Obi OS
06:30 AM	Generator Operator	Check generator oil, water levels; start generator; record diesel dip.	Kenneth logs starting diesel dip.
07:00 AM	Supervisor (Kenneth)	Open raw water supply; backwash sand & carbon filters; test TDS and pH.	Kenneth logs TDS (24 ppm) and pH (7.2).
07:30 AM	Packaging Staff	Mount nylon rolls on sachet machines; load PET preforms into blow mold heater.	Kenneth unlocks console (PIN 1111).
08:00 AM	Loading Bay & Drivers	Inspect delivery trucks; load mixed orders for morning route dispatch.	Musa inspects loading manifest (PIN 3333).
08:30 AM	Gate Cashier (Blessing)	Open sales kiosk; count starting drawer cash; test Moniepoint POS terminal.	Blessing unlocks POS (PIN 2222).
12:30 PM	Midday Shift Audit	Inspect blow-mold rejects; verify sachet machine counter against finished bags.	Kenneth weighs crushed scrap plastic.
05:30 PM	Trucks Return	Drivers return; gate loaders count empty sachets and 19L dispenser jars.	Musa inputs returned jars & burst pouches.
06:00 PM	Cashier Cashing Up	Blessing balances drawer cash against sales register; settles Moniepoint receipts.	Blessing confirms cash and transfer totals.
06:30 PM	Supervisor Shift Close	Kenneth records final diesel dip; taps 1-Tap WhatsApp Closing Trigger.	WhatsApp digest sent to Mr. Obi.
08:00 PM	Mr. Obi (Owner Audit)	Mr. Obi opens owner portal on phone; audits net cash profit (₦524,200).	Full plant reconciled & closed.

5.2 Preventive Machine Maintenance Schedule

Daily Maintenance Checklist

- Clean vertical and horizontal heat-sealing jaws on sachet machines.
- Inspect Teflon sealing tape for burns or peeling; replace if torn.
- Blow mold air compressor: drain water condensation from receiver tank.
- Wipe down 19L jar sanitizing chamber and bottle conveyor tracks.

Weekly & Monthly Maintenance

- Backwash multi-media sand and activated carbon filters (every 48 hrs).
- Replace 5-micron and 1-micron spun sediment pre-filter cartridges (monthly).
- Clean RO membrane with food-grade citric acid wash if TDS rises > 50 ppm.
- Service generator: oil change, fuel filter and air filter (every 250 hrs).

6. Emergency Troubleshooting, Offline Recovery & Support

6.1 Troubleshooting Common Factory Failures

Failure Scenario	Root Cause	Immediate Physical Remedy	System Action on Obi OS
Sachet pouches leaking at bottom seal	Horizontal sealing jaw temperature too low or Teflon tape worn out.	Increase temperature controller by 5°C; replace burnt Teflon tape strip.	Kenneth logs any burst pouches during test runs into machine waste.
Blown PET bottles warping or cracking	Infrared preform heating lamps uneven or air compressor pressure < 25 bar.	Adjust preform rotation speed in heating oven; verify high-pressure compressor.	Kenneth enters damaged bottles into Preform Scrap Grammage Audit.
TDS meter reading rises above 70 ppm	Borehole salt surge or RO membrane scaling / fouling.	Stop bottling line immediately; run chemical CIP membrane wash with citric acid.	Log TDS in NAFDAC log; withhold production until TDS drops < 45 ppm.
Phone battery dies mid-shift	Factory phone depleted during active production logging.	Plug phone into generator charger. Open browser tab once powered.	**Offline Ledger hydrates state instantly with zero data loss.**
Internet drops (MTN / Airtel offline)	Cellular mast downtime or data bundle exhausted.	Continue operations normally. System functions 100% offline via LocalStorage.	Data caches locally and syncs to cloud automatically upon reconnection.

6.2 Offline Disaster Recovery Protocol

> **Zero Data Loss Guarantee:** The Obi Table Water OS is built with an **Offline-First Persistence Ledger** (`src/lib/factory/offlineLedger.ts`). Every button tap, sale entry, preform weight, and diesel volume is immediately committed to the phone's non-volatile local storage. Even if the phone shuts down, browser crashes, or the cellular data network fails for 6 hours, re-opening the application restores the active shift state to the exact second.

6.3 Technical Support & Operational Hotline

Bethelmind Digital Solutions (CAC Certified Enterprise: Reg 9912717)

For technical support, custom software modifications, training consultations, or emergency cloud server recovery:

- **Technical Desk Hotline:** 0802 279 1227
- **WhatsApp Executive Support:** wa.me/2348022791227
- **Email Support:** bethelmindrecruit@gmail.com
- **Corporate Bank:** Moniepoint Microfinance Bank
- **Settlement Account:** 6805375225 (Bethelmind Digital Solutions)
- **Live Portal URL:** https://www.bethelmindanalytics.com/factory